

How robust are traditional stock price models under bubble-like
market dynamics?
Analysis of the AI bubble

Mya Mokrane
McGill University
261104994

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Abstract

The AI bubble.

Introduction

The rapid development of artificial intelligence has had not only drastic impacts on our daily lives and future careers, but also a significant impact on many financial markets and enterprises. Indeed, AI has not only become a tool many of us use and rely on for research, and advice for individuals and governments but also a revolutionary tool in scientific experiments, models, and analysis. As a result, the investments and the development of such a tool has become a pivotal and global trajectory for the future of many fields. In response to the extraordinary growth in demand for AI services and the associated chiptechnology, AI-focused companies have experienced soaring stock prices, which have raised concerns about the potential formation of speculative bubbles.

Major technology companies such as NVIDIA, META, or Microsoft have thus had a major increase in their stock prices in the past decade.

This can lead to the following question : how relevant are traditional financial models in predicting AI stock prices ? The Black-Scholes model, developed in 1973, presented a widely used option pricing formula.

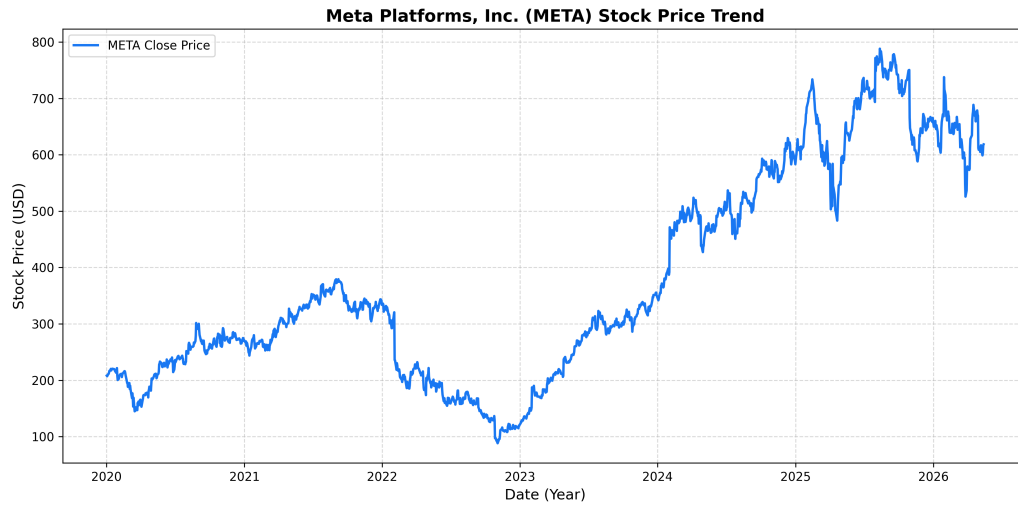


Figure 1: META closing stock price since 2020.

As shown in Figure 2, META experienced substantial volatility during the post-pandemic AI expansion period.

Black-Scholes Equation

In 1973, Fisher Black and Myron Scholes published their groundbreaking work on option pricing. Indeed, in their article "The Pricing of Options and Corporate Liabilities" [BS73], they derived the Black-Scholes formula for option pricing:

$$C^{BS}(t, S; r, \sigma, K, T) := S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2) \quad (1)$$

where Φ denotes the standard normal distribution (cdf), r represents the continuously compounded risk-free interest rate, σ denotes the volatility of the underlying stock, and where

$$d_1 = \frac{\ln \frac{S}{K} + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

We will use standard market practices where time is measured in years and r and σ are expressed in annualized terms.

Furthermore, we will set S_t to be the price of the stock at time t . Real market interest rates and volatilities change constantly and thus depend on time : r_t, σ_t . As a result, we take the vector of risk-factors as a function of time: $\mathbf{Z}_t = (\ln S_t, r_t, \sigma_t)'$.

The risk-factor changes are given by :

$$\mathbf{X}_{t+1} = (\ln S_{t+1} - \ln S_t, r_{t+1} - r_t, \sigma_{t+1} - \sigma_t)'. \quad (2)$$

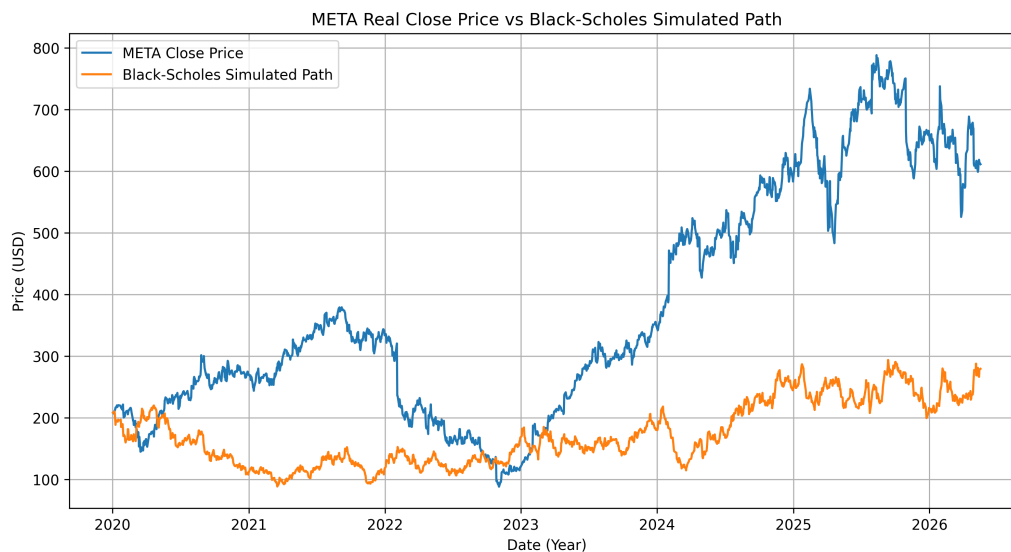


Figure 2: META closing stock price vs Black Scholes Simulated path since 2020.

New Methods

Times Series Analysis : GARCH Model

AI Stress-Testing

References

- [BS73] Fischer Black and Myron Scholes. “The pricing of options and corporate liabilities”. In: *Journal of political economy* 81.3 (1973), pp. 637–654.